# Where does weight-space programming fix failures?

NVIDIA Cosmos3 Edge Policy (world-action model) and NVIDIA GR00T N1.7 DROID (VLA on a Cosmos-Reason2-2B backbone), with OpenPI π0.5 as the control, on the same RoboLab DROID tasks Cube then banana, Eyeglasses, Cube left of bowl, Yellow on red, Mustard in left bin, Three-cube tower, Flat box, Mustard right of raisin box and a retention task Cube in bowl.

**Method.** When a clearer wording of the same goal makes a robot succeed, copy that wording's internal effect into the weights: one least-squares calculation per layer, no training, and the robot still gets the original instruction.

**π0.5, the control (earlier records).** The edit fixed its two wording failures: Cube then banana 27 → 106 of 128 unseen starts and Mustard right of raisin box 0 → 27 of 64; a random edit of the same size did nothing (9 vs 13 of 64); no edit helped the tasks that fail for other reasons.

**NVIDIA models. The fix:** On NVIDIA's Cosmos3 Edge, a weight edit fixed a task the robot failed only because of how its instruction was worded: 0 → 56 successes from 64 unseen starts. A random edit of the same size, a placebo: 0. **Where it works:** Only when written into the part of the model that reads the instruction. **The cost:** It breaks the two other tasks whose instructions say "left": Cube left of bowl 13 → 0 and Mustard in left bin 9 → 0 of 32 unseen starts.



Cosmos3 Edge Policy, Mustard right of raisin box: the wording gap; the clip plays on the live page.

### What weight-space programming did on each task, on unseen starts unless marked practice

| | Cosmos3 Edge Policy<br>world-action model | GR00T N1.7<br>reasoning-backbone VLA | π0.5<br>OpenPI VLA, control |
|---|---|---|---|
| Cube then banana | Mustard right of raisin box edit: 18 → 18 of 32<br>9 / 9; p = 1 | 9 → 0 of 32 practice starts<br>0 / 9, p = 0.0039 | fixed: 27 → 106 of 128 earlier slots<br>distilled edit; 81 / 2, p = $7.2 \times 10^{-22}$; shared slots, unedited: 5 of 16 |
| Eyeglasses | Mustard right of raisin box edit: 4 → 1 of 16<br>0 / 3; p = 0.25 | unedited 0 of 32<br>not edited | Cube then banana edit: 22 → 44 of 64 earlier slots<br>own edit 18 → 27 of 64, not significant |
| Cube left of bowl | Mustard right of raisin box edit: 13 → 0 of 32<br>0 / 13; p = $2.4 \times 10^{-4}$ | unedited 1 of 32<br>not edited | Cube then banana edit: 8 → 36 of 64 earlier slots<br>no own edit submitted |
| Yellow on red | Mustard right of raisin box edit: 24 → 22 of 32<br>4 / 6; p = 0.75 | unedited 0 of 16<br>not edited | no gain earlier slots<br>lookalike perception; no clear wording |
| Mustard in left bin | Mustard right of raisin box edit: 9 → 0 of 32<br>0 / 9; p = 0.0039 | unedited 0 of 32<br>not edited | no gain earlier slots<br>own edit 8 → 11 of 64; the grasp slips |
| Three-cube tower | Mustard right of raisin box edit: 1 → 1 of 32<br>1 / 1; p = 1 | unedited 0 of 16<br>not edited | no gain earlier slots<br>own edit 35 → 39 of 64 |
| Flat box | Mustard right of raisin box edit: 0 → 0 of 16 practice starts<br>0 / 0; p = 1 | unedited 0 of 32<br>not edited | no gain earlier slots<br>a missing skill; 0 of 64 unedited |
| Mustard right of raisin box | 0 → 56 of 64<br>56 / 0, p = $2.8 \times 10^{-17}$ | unedited 0 of 32<br>not edited | fixed: 0 → 27 of 64 earlier slots<br>distilled edit; 27 / 0, p = $1.5 \times 10^{-8}$ |
| Cube in bowl | 24 → 28 of 32 | unedited 28 of 32<br>not edited | kept: 32 → 32 of 32 earlier slots<br>cube into the bowl, Cube then banana edit |
| Rendering: performance → default preset | *running* | 5 → 16 of 16<br>Banana onto plate | *running* |
| Random control, same size | control 0, unedited 0, edit 56 of 64<br>Mustard right of raisin box | control 11, unedited 9, edit 0 of 32 practice starts<br>Cube then banana | 9 of 64 earlier slots<br>Cube then banana; unedited 13, edit 50 |
| Where to edit | text pathway 13, video-action (generation) pathway 0 of 16<br>Mustard right of raisin box, 16 practice starts; unedited 0 | Cosmos-Reason2 backbone 0, diffusion action head 0 of 32<br>Cube then banana, 32 practice starts; unedited 9 | VLM 55, action expert 20, head 0 of 64 earlier slots<br>Cube then banana; unedited 5; only the VLM edit transfers |

| | Cosmos3 Edge | GR00T N1.7 | π0.5 |
|---|---|---|---|
| Cube then banana | 10/16 · 63% | 9/32 · 28% | 5/16 · 31% |
| Eyeglasses | 0/16 · 0% | 0/32 · 0% | 18/64 · 28% |
| Cube left of bowl | 4/16 · 25% | 1/32 · 3% | 8/64 · 13% |
| Yellow on red | 12/16 · 75% | 0/16 · 0% | 13/32 · 41% |
| Mustard in left bin | 8/16 · 50% | 0/32 · 0% | 8/64 · 13% |
| Three-cube tower | 1/16 · 6% | 0/16 · 0% | 35/64 · 55% |
| Flat box | 0/16 · 0% | 0/32 · 0% | 0/32 · 0% |
| Mustard right of raisin box | 0/16 · 0% | 0/32 · 0% | 0/64 · 0% |
| Cube in bowl | 9/16 · 56% | 28/32 · 88% | 32/32 · 100% |

Unedited success on each shared task; stronger shading is higher. Each cell gives successes of starts.

| | Cosmos3 Edge | GR00T N1.7 | π0.5 |
|---|---|---|---|
| Cube then banana | ±0 / 32 | **−9 / 32*** | **+79 / 128*** |
| Eyeglasses | −3 / 16 | not run | **+22 / 64*** |
| Cube left of bowl | **−13 / 32*** | not run | **+28 / 64*** |
| Yellow on red | −2 / 32 | not run | +3 / 32 |
| Mustard in left bin | **−9 / 32*** | not run | +5 / 32 |
| Three-cube tower | ±0 / 32 | not run | +5 / 32 |
| Flat box | ±0 / 16 | not run | ±0 / 32 |
| Mustard right of raisin box | **+56 / 64*** | not run | not run |
| Cube in bowl | +4 / 32 | not run | ±0 / 32 |

gain, p < 0.05 · loss, p < 0.05 · no detectable change

* p < 0.05; outlined: the edit's own task

Each model's main edit, unchanged, on every suite task: edited minus unedited successes on the same starts. Teal: a gain; near-black (near-white in dark mode): a loss (exact McNemar p < 0.05); grey: no detectable change; the outlined cell is the task the edit was built for.

Live page with every table, figure and record: ali-adeeb.com/weightsmith/cosmos · Paper: ali-adeeb.com/weightsmith/paper.pdf · Code and records: github.com/adeeb10abbas/weightsmith

Draft for discussion. Simulation only (RoboLab, Isaac Sim), one released checkpoint per model, shared slots 30000–30031 and 96000–96063 with identical initial scenes; π0.5 cells are earlier records on their own slots until its shared-slot runs land. Exact McNemar tests on paired starts.